

AI-Enabled Smart Electric Vehicle Charging and Energy Management Platform

Sub-Project 1 —Phase 1 Report

Data acquisition, preprocessing and demand-forecasting baseline

Sub-project	SP1 —EV charging data analysis and demand forecasting
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Supervisor	Dr Amer Ghias (University of Glasgow / Glasgow College UESTC)
Date	14 September 2026
Deliverables covered	Clean datasets; analysis framework; forecasting models; accuracy evaluation
Code	subprojects/sp1-data-forecasting/
Outputs	data/processed/sp1-demand-forecast-v1.csv data/processed/sp1-baseline-demand-v1.csv

1 Scope and objectives

SP1 owns the first stage of the platform’s data flow: it turns raw public charging records into the gap-free hourly demand series that every other sub-project consumes (SP2 schedules against it, SP4 displays it, SP5 prices it). The official brief assigns SP1 four tasks: collect and preprocess public EV charging datasets; analyse charging patterns and user behaviour; develop machine-learning models to predict future charging demand; and evaluate forecasting accuracy and model performance. This report covers all four, and with them the Phase-1 milestone (datasets secured, tools aligned, framework running).

Two questions shaped the work and are answered in Section 2:

1. **Which data?** The supervisor’s kickoff email named three sources —Caltech ACN-Data, Boulder Colorado, and UK National Grid. All three were checked against their live services, and the outcome decided the project’s data foundation.
2. **Is it official?** The team’s earlier plan assumed Chinese datasets. A member-account review of the Chi-na Charging Alliance platform and a search of provincial and grid-operator portals found no session-level Chinese data at all, so that route could not carry a forecasting model.

2 Data acquisition and source evaluation

2.1 Sources verified

Every source below was probed with live requests (HTTP status, schema, row counts, licence text); nothing is quoted from a landing page without having been observed. The full verification log is kept in

the repository at `data/raw/official-datasets-research.md`.

Source	Official?	EV charging data?	Access observed	Verdict
City of Boulder, Colorado	Yes — municipal government	Yes — sessions, metered kWh	ArcGIS FeatureServer, no registration; 148,136 rows pulled	In use (primary) CC0 1.0
City of Palo Alto, California	Yes — municipal government	Yes — sessions, metered kWh, user IDs	Direct CSV, no registration; 85.4 MB, 259,415 rows	In use (validation) PDDL
Caltech ACN-Data	Yes — Caltech + PowerFlex	Yes — sessions <i>plus</i> measured power time series	API returns HTTP 401; free self-registration issues a token	Obtainable; not used yet (see §6)
UK National Grid (now NESO)	Yes — national system operator	No — GB national demand only	CKAN/CSV, no registration, 2001–2025	Not an EV dataset; grid context only
China Charging Alliance	Yes — industry alliance	No — monthly aggregates only	Member account required	Could not support SP1

Table 1: Sources checked for SP1. Only the first two provide session-level EV charging records that are official *and* downloadable without a signed agreement.

Two consequences follow, and both are recorded in the integration contract as amendment A1 rather than being decided quietly:

- **Region.** The project can no longer be described as China-only; the demand series is for two US cities. The file formats are unchanged.
- **Currency.** Neither dataset contains a tariff or price column, so CNY pricing has no basis. SP2 must source a US tariff schedule before it can populate `tariff_cny_per_kwh`; that is now an open item owned by SP2.

2.2 Datasets used

Property	City of Boulder (primary)	City of Palo Alto (validation)
Publisher	City of Boulder Open Data	City of Palo Alto Open Data
Licence	CC0 1.0 (public domain)	PDDL (public domain)
Sessions	148,136	259,415
Stations	50 (all Level 2)	47
Period	2018-01-02 – 2023-12-01 (5.9 years)	2011-07-30 – 2021-01-01 (9.4 years)
Energy column	Energy (kWh) — metered	Energy (kWh) — metered
Timestamps	start and end, minute resolution	start and end, minute resolution
Per-vehicle ID	none	User ID, 97% populated
Hourly series	<i>derived</i> (see §3)	<i>derived</i>
Rows with zero energy	16,113 (10.9%)	—

Table 2: The two datasets actually used. Both are ChargePoint-style exports, so both share the same traps, handled below.

2.3 Cleaning decisions

The datasets are published as session records, so “cleaning” here means mapping them into one canonical schema and refusing to let known traps through silently.

1. **Occupancy is not charging time.** Both sources publish a total plugged-in duration *and* an active charging duration. In Boulder the median occupancy is 1.55 h but the tail reaches 839 h (vehicles left connected for weeks). Spreading a session’s energy across occupancy would smear the load profile into nonsense, so the adapter maps the canonical duration to *charging* time (median 1.39 h, 99th percentile 8.3 h) and defines the session window as `start → start + charging_time`.
2. **Units and types.** Boulder delivers every column as a string, including numerics; the adapter casts explicitly and reports unparsable values instead of dropping rows. Durations arrive as `hh:mm:ss` rather than minutes, which the adapter now parses in either form.
3. **Timezone.** Storage is UTC. Source timestamps are naive local time; each dataset declares its own zone (`America/Denver`, `America/Los_Angeles`) and is converted on load, so daylight saving is handled by the timezone database rather than by the file’s own (sometimes inconsistent) zone label.
4. **Energy provenance.** These datasets measure delivered energy. Where a dataset instead publishes only a *rated* power, energy becomes a derived upper bound; that is a materially weaker input and was one reason the Chinese third-party dataset was set aside.

3 Method

3.1 From sessions to an hourly demand series

Each session contributes its metered energy to every hourly interval it overlaps, in proportion to the overlap duration. The result is energy per interval, which at a one-hour resolution is numerically the average power in kW. Absent intervals mean “no charging recorded”, which for demand is a true zero, so the grid is completed with zeros; the contract’s guarantee to downstream sub-projects is a *gap-free, non-negative, UTC-stamped* series, and the pipeline re-reads and validates every file it writes.

The one modelling assumption worth stating plainly: the datasets contain no intra-session power trace, so a session’s energy is spread *evenly* across its charging window. Real charging curves taper, so this slightly smooths peaks within a session. The assumption is recorded in each output’s `.meta.json` under `interval_method` so no downstream sub-project has to guess.

3.2 The uncontrolled baseline

SP2 and SP5 need something to compare an optimised schedule against, so SP1 also emits an “uncontrolled” profile: every session’s energy is moved to a block starting at 18:00 local on the day it occurred, charged at that session’s own recorded power. Energy is therefore conserved and only its timing changes—a deliberate worst-case shape, not a network simulation. (At a fixed 7 kW this would stretch a 150 kW fast-charge session over 40 hours; using each session’s own power avoids that artefact.)

3.3 Features and forecasting

Forecasting is framed as *direct multi-horizon forecasting with the horizon as a feature*: one model predicts all 24 hours from a single origin, so errors do not accumulate the way they do in recursive one-step prediction.

- **Known-future features** describe the *target* interval: local hour of day, day of week, weekend flag, month, and cyclical encodings. These are known in advance, so using them is legitimate.
- **Autoregressive features** describe the *origin*: demand lags of 1, 2, 3, 24, 48 and 168 hours and rolling means, standard deviations and maxima over 24 and 168 hours, all shifted so they cannot see the interval being predicted.
- **Naive references** are first-class inputs: the value 24 hours and 168 hours before the target, which also serve as the persistence baselines.

A regression test recomputes every lag from the raw series and fails if any feature can see its own target; a second one pins the training boundary with a ramp series, because a leaky backtest reports fantasy accuracy.

3.4 Models and evaluation

Seven models are scored: three baselines (`naive_daily`, `naive_weekly`, `hour_of_week_profile`) and four learned models (ridge regression, random forest, gradient boosting, XGBoost). Evaluation is a **rolling-origin backtest**: for each origin the model is retrained using only targets strictly before that origin, and forecasts the next 24 hours. Origins are anchored to local midnight, so each is a genuine day-ahead forecast, and only origins with a full horizon are used, which keeps the evaluation window gap-free. Training history per origin is capped at 180 days.

Accuracy is reported as **WAPE** (total absolute error divided by total demand), which stays stable when many intervals have near-zero demand, alongside MAE, RMSE and R^2 . Confidence bounds are empirical: $\hat{y} \pm 1.645 \sigma$ where σ is the residual standard deviation at the same horizon in the backtest, clipped at zero.

4 Results

4.1 Charging patterns

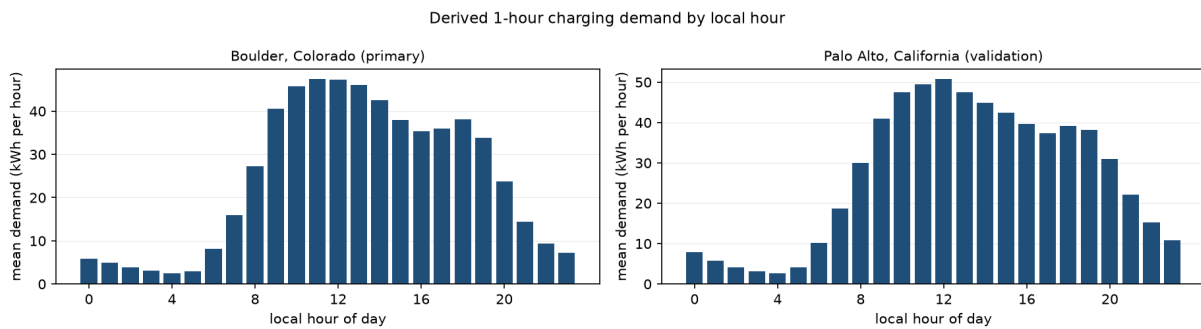


Figure 1: Mean derived demand by local hour of day. Both sites are Level-2 workplace and municipal charging, so both peak in the middle of the working day rather than in the evening —the residential 18:00 peak of the project brief does not appear in either dataset.

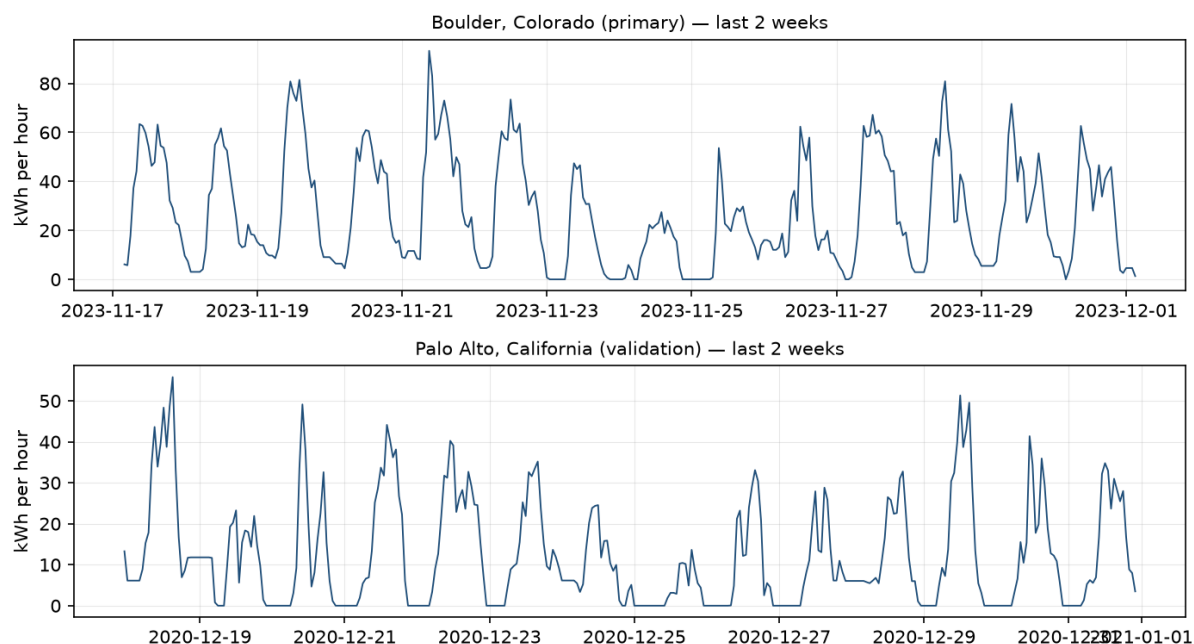


Figure 2: The last two weeks of each hourly series, in local time: a strong daily cycle with visibly lower weekend activity.

Boulder’s 148,136 sessions aggregate to 1.253 GWh over 5.9 years (mean 0.58 MWh per day). The local-hour profile (Figure 1) is unmistakably a workplace fleet: demand climbs from 06:00, peaks between 10:00 and 13:00 (the 11:00 hour alone carries 8.2% of daily energy), and decays after 20:00 to a 04:00 trough (0.42%). The peak-to-trough ratio is 19.3 $\times$, so the *timing* of charging matters greatly even at this small scale. A secondary bump at 17:00–18:00 (6.2–6.6%) is consistent with vehicles plugging in at the end of the working day. Palo Alto’s 259,415 sessions give 2.216 GWh over 9.4 years (mean 0.64 MWh per day) with a strikingly similar shape: peak at 11:00–13:00 (7.4–7.9%) and a 04:00 trough (0.41%), peak-to-trough 19.2 $\times$. Two independently published municipal networks showing the same daily signature is a useful check on the preprocessing: the shape is a property of Level-2 workplace charging, not of one city’s idiosyncrasies. It also means the project brief’s evening residential peak is *absent* from both datasets, which the scenario design in Phase 3 must acknowledge rather than assume.

4.2 Forecast accuracy

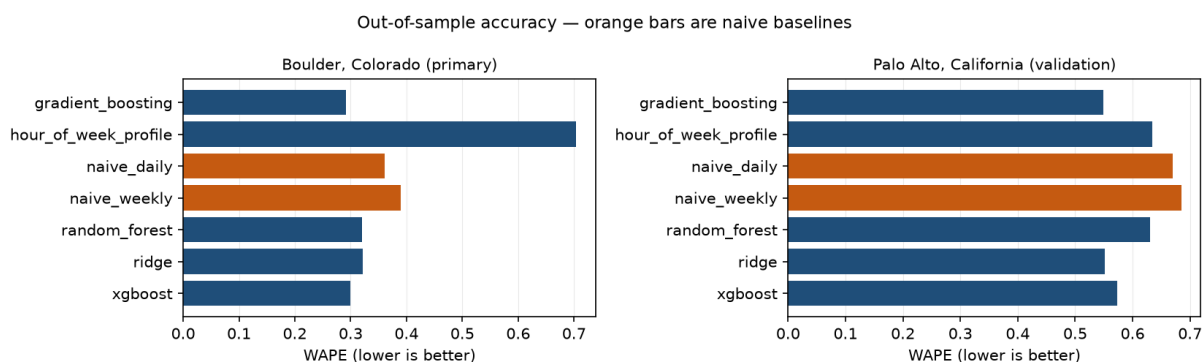


Figure 3: Out-of-sample WAPE by model. Orange bars are the naive baselines; a learned model only earns its complexity by beating them.

Boulder (primary), 25 daily origins $\times$ 24 hours = 600 out-of-sample intervals. Gradient boosting is the best model; every learned model beats both naive baselines, and the ranking is stable across the horizon (per-horizon WAPE 0.196–0.910).

Model	WAPE	MAE (kWh/h)	RMSE (kWh/h)	R^2
gradient boosting	0.2920	8.37	11.39	0.763
XGBoost	0.2999	8.60	11.79	0.747
random forest	0.3197	9.17	12.53	0.714
ridge regression	0.3217	9.23	12.25	0.727
naive daily	0.3602	10.33	13.97	0.644
naive weekly	0.3891	11.16	15.45	0.565
hour-of-week profile	0.7040	20.19	25.76	−0.210

Table 3: Boulder: out-of-sample accuracy, rolling origin, retrained per origin.

Gradient boosting improves on daily persistence by **19% in relative WAPE** (0.292 against 0.360), which is the result that matters: on this data the learned model *earns its complexity*. The seasonal-profile baseline is worse than useless ($R^2 < 0$), because averaging by weekday and hour cannot track the autumn drift visible in Figure 2. **Palo Alto (validation), 14 daily origins = 336 out-of-sample intervals.** The same model wins, and the learned models again beat persistence — but absolute accuracy is markedly worse, which the evaluation window explains: it ends on 2020-12-31, i.e. inside the COVID-19 period when workplace commuting collapsed. This is a limitation of the data, not of the method, and it is reported rather than tuned away.

Model	WAPE	MAE (kWh/h)	RMSE (kWh/h)	R^2
gradient boosting	0.5491	7.02	9.70	0.466
ridge regression	0.5514	7.05	9.79	0.456
XGBoost	0.5731	7.33	9.98	0.434
random forest	0.6301	8.06	10.84	0.333
hour-of-week profile	0.6341	8.11	11.17	0.292
naive daily	0.6702	8.57	11.74	0.217
naive weekly	0.6846	8.75	12.63	0.094

Table 4: Palo Alto: out-of-sample accuracy on a disrupted (COVID-era) window.

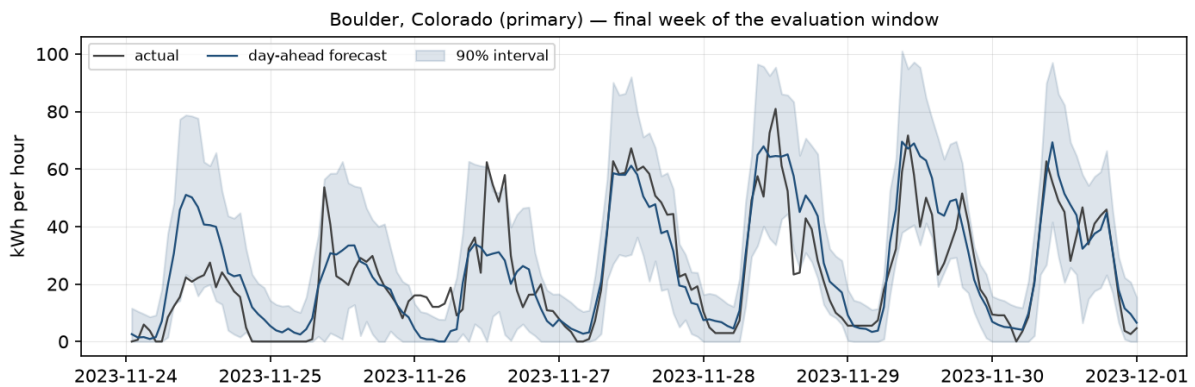


Figure 4: Final week of the Boulder evaluation window: the day-ahead forecast against the realised demand, with the empirical 90% interval.

4.3 Uncontrolled baseline versus forecast

The second contract file moves every session to an 18:00 local start at that session’s own power, which conserves energy and changes only its timing. On Boulder’s 600-hour window it turns a 94.5 kWh peak into a **362.5 kWh peak** — a $3.8\times$ **amplification** while total energy is preserved to within 0.05% of the realised demand. That factor is the concrete, measurable version of the brief’s “50 EVs plug in at 18:00” narrative, and it is what SP2’s scheduler exists to remove.

Quantity (Boulder, 600 h)	Forecast	Uncontrolled baseline
Total energy	17.81 MWh	17.22 MWh (realised: 17.21 MWh)
Peak hour	94.5 kWh	362.5 kWh
Mean load	29.7 kWh/h	28.7 kWh/h
Energy bias vs realised	+3.5%	+0.05%
90% interval (mean width)	33.2 kWh	not modelled (empty cells)

Table 5: The two contract deliverables over the same window.

One further check worth reporting: the empirical coverage of the 90% interval is **91.0%** on Boulder and 90.5% on Palo Alto. The residual-based interval is therefore calibrated, not decorative — the intervals are wide (116% of mean demand on Boulder), but they mean what they claim.

5 Limitations and threats to validity

1. **The hourly curve is derived, not measured.** Both datasets are session-level; no native hourly or sub-hourly power series exists in either. The derivation is documented and its plausibility was checked (Palo Alto's implied average power has a median of 3.7 kW, consistent with Level-2 AC charging), but a within-session taper is not represented.
2. **Level-2 AC only, small fleets.** Boulder has 50 stations, Palo Alto 47. Both are workplace/municipal AC charging, so the brief's "50 EVs plug in at 18:00" scenario must be constructed as a stated scenario rather than read off the data. The grid-stress story is about *when* energy is drawn, which these datasets do support.
3. **Region mismatch with the brief.** The original framing was a Chinese market project. The data are US municipal. This was a deliberate trade of market relevance for data authenticity and licensing, and it needs the supervisor's acknowledgement.
4. **No tariff data**, so SP1 cannot produce cost; that is SP2's input to source.
5. **External shocks.** Palo Alto's record runs to the end of 2020, so its final evaluation window sits in the COVID-19 period, when commuting and therefore workplace charging collapsed. Any comparison between the two sites must not treat that as a modelling failure.
6. **Short evaluation windows.** A few dozen daily origins per site at a 24-hour horizon give a stable but not large sample; the ranking should be re-checked as more history accumulates.

6 Phase-2 plan

1. **Answer the aggregate-vs-per-EV question properly.** Palo Alto carries a user identifier for 97% of sessions, so a per-EV forecast is now a scope decision rather than a data limitation. This is the interface question that still blocks SP2's optimiser design.
2. **Consider Caltech ACN-Data** as a third site: it is the only source reviewed that publishes measured power time series, which would let SP1 test how much the even-spreading assumption costs. It needs a free research registration.
3. **Source a US tariff schedule** (open item for SP2) so the economic chain $SP2 \rightarrow SP5$ can be populated.
4. **Extend the model comparison** with the mid-project milestones in view: longer training windows, a per-site model, and an explicit test of whether the learned models' advantage over persistence survives

at longer horizons.

5. **Literature review** —the SP1-relevant strand (demand forecasting for EV charging, and the standard practice of session-to-load disaggregation) remains to be written up; this report deliberately does not claim to have completed it.

Reproducibility

```
python -m venv .venv && .venv\Scripts\activate
pip install -r requirements.txt
```

```
# data (never committed; both fetch scripts verify what they download)
python subprojects/sp1-data-forecasting/scripts/fetch_datasets.py --dataset us-boulder-ev
python subprojects/sp1-data-forecasting/scripts/fetch_datasets.py --dataset us-palo-alto-ev
```

```
# analysis + contract outputs
python subprojects/sp1-data-forecasting/scripts/run_sp1.py --dataset us-boulder-ev \
    --forecast-days 30 --train-window-days 180
python subprojects/sp1-data-forecasting/scripts/run_sp1.py --dataset us-palo-alto-ev \
    --forecast-days 14 --train-window-days 180
```

```
# figures and tests
python subprojects/sp1-data-forecasting/scripts/make_figures.py
python -m pytest subprojects/sp1-data-forecasting/tests -q
```